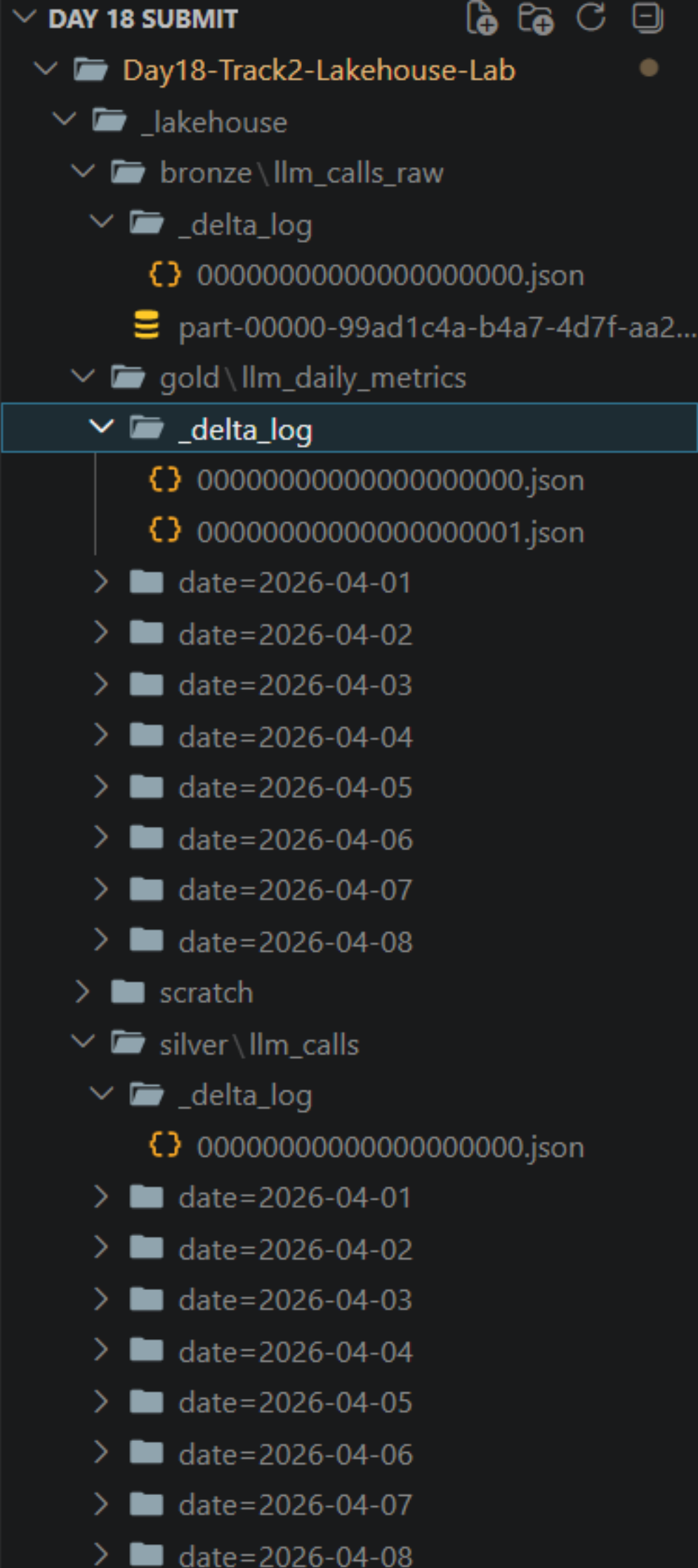




3. Schema enforcement — try to write a wrong schema

```
bad = pl.DataFrame({"id": [4], "name": ["dan"], "age": ["thirty"], "city": ["Hue"]})
try:
    write_deltalake(table_path, bad.to_arrow(), mode="append")
    print("UNEXPECTED: bad write succeeded — schema enforcement broken")
except Exception as e:
    msg = str(e).splitlines()[0][:120]
    print(f"BLOCKED by schema enforcement (expected): {type(e).__name__}: {msg}")
```

BLOCKED by schema enforcement (expected): Exception: Cast error: Cannot cast string 'thirty' to value of Int64 type

1. `df` is a `pandas.DataFrame` object.

shape: (4, 5)

id	name	age	city	tier
---	---	---	---	---
i64	str	i64	str	str
1	alice	30	Hanoi	null
2	bob	25	HCMC	null
3	charlie	35	Danang	null
4	dan	28	Hue	premium

Files before OPTIMIZE: 200

—— Z-order deliverable metrics ——

Speedup (wall-clock): 8.3× (target $\geq$ 3×)

Files-pruned ratio: 55.0× (target $\geq$ 10×) [1 of 55 files cover user_id=4242]

AFTER OPTIMIZE+ZORDER count=5 median= 15.3 ms (n=3)

Speedup: 8.3x (target $\geq$ 3x)

File reduction: 200 $\rightarrow$ 55 (4x fewer)

v 4 RESTORE

v 3 WRITE

v 2 MERGE

v 1 WRITE

v 0 WRITE

Total versions: 5 (target ≥ 5)

MERGE 100K rows: 0.28s

RESTORE → v2: 0.02s (target < 30s)

Rows with score<0 after restore: 0 (expected 0)

▼  Day18-Track2-Lakehouse-Lab

▼  _lakehouse

>  bronze

>  gold

>  scratch

>  silver

Silver rows: 190,052 (Bronze 200,000 → dedup dropped 9,948)

shape: (24, 8)

date	model	p50_latency_ms	p95_latency_ms	total_prompt_tokens	total_completion_tokens	error_rate	cost_usd
---	---	---	---	ns	---	---	---
date	str	f64	f64	---	---	f64	f64
				decimal[38,0]	decimal[38,0]		
2026-04-02	claude-haiku-4-5	565.0	1140.0	16651653	8245961	0.049327	46.305166
2026-04-02	claude-opus-4-7	3004.0	5972.0	5579305	2767949	0.052803	291.28575
2026-04-02	claude-sonnet-4-6	1388.0	2749.0	33145603	16392052	0.049926	345.317589
2026-04-03	claude-haiku-4-5	570.0	1136.8	16109287	8034124	0.047066	45.023926
2026-04-03	claude-opus-4-7	3037.0	5976.65	5601091	2840460	0.043369	297.050865
...	...	...	...	...	...	...	...
2026-04-07	claude-opus-4-7	3069.0	5967.0	5442975	2713496	0.061844	285.156825
2026-04-07	claude-sonnet-4-6	1391.0	2748.85	32464006	16377557	0.048627	343.055373
2026-04-08	claude-haiku-4-5	574.0	1118.3	4855783	2403003	0.042171	13.496638
2026-04-08	claude-opus-4-7	2945.5	6058.05	1571994	760101	0.059367	80.587485
2026-04-08	claude-sonnet-4-6	1404.0	2782.85	9596279	4882116	0.049979	102.020577

— Gold deliverable metrics —
Distinct dates: 8 (target ≥ 7)
Distinct models: 3
Total Gold rows: 24 (= dates × models)